

Q1: ANSWER THE FOLLOWING QUESTIONS:

1. List and Draw the Knowledge Discovery (KDD) Process steps. ✓

2. Data in the real world is dirty. Discuss this statement. ✓

3. List different qualitative data types with example ✓

4. List different Data cleaning tasks ✓

5. There are various reasons for missing values. Discuss how to handle missing values ✓

6. Define the main Apriori principle and the two main steps of this Algorithm. ✓

7. What is the difference between Maximal Frequent Itemset and Closed Itemset

Q2- Specify the suitable functionality of data mining for each of the following tasks. If it is not a data mining task choose "e"

a. Clustering	b. association analysis
c. Classification	d. Regression
e. Not data mining	

1- Predicting the future stock price in dollars of a company using historical records. C B D

2- Computing the total sales of a company. Not mining E

3- classify countries based on (climate) A C

4- cluster houses to find distribution patterns. A

5- What items are frequently purchased together in your B

Q3- Classify the following attributes as binary, discrete, or continuous.

Also classify them as qualitative (nominal or ordinal) or quantitative (interval or ratio).

1. Height above sea level Quant, Ratio, Contin.

2. Military rank. ordinal, contin, quali

3. Type of a portrait discrete, nominal, qualit

4. Movie rating ordinal, qualitative

5. Number of correct answers on a quiz quantit, interval, continuous

Q4- List and define five Classifier Evaluation Metrics. Then, Using the following confusion matrix calculate the defined measures.

Predicted Class \ Actual Class	cancer = yes	cancer = no	Total
cancer = yes	90	210	300
cancer = no	140	9560	9700
Total	230	9770	10000

	yes	No
yes	TP	FN
No	FP	TN

$S \rightarrow 30$

$S \rightarrow 98.5$

$A \rightarrow 96.5$

$E \rightarrow 0.0351$

$P \rightarrow 39$

$R \rightarrow 30$

$F \rightarrow 881.739$

With my best wishes
Dr Ayatallah Gamal